SIMATS ENGINEERING

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Machine Learning

Experiment -9

Compare Linear and Polynomial Regression using Python

Code:

```
# Compare Linear and Polynomial Regression

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
df = pd.read_csv("play_tennis (1).csv")

# Select a single numerical feature for comparison and plotting
# The 'day' column will be converted to a numerical feature.
# The 'play' column will be converted to a numerical target (0 for No, 1 for
Yes).
X = df[["day"]].copy() # Use .copy() to avoid SettingWithCopyWarning
y = df["play"].copy()

# Convert 'day' column to numerical (e.g., D1 -> 1, D2 -> 2)
X['day'] = X['day'].str[1:].astype(int)

# Convert 'play' column to numerical (No -> 0, Yes -> 1)
y = y.map({'Yes': 1, 'No': 0})

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# -------------------------------
# Linear Regression
# -------------------------------

linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

linear_pred = linear_model.predict(X_test)

linear_mse = mean_squared_error(y_test, linear_pred)
linear_r2 = r2_score(y_test, linear_pred)

# -------------------------------
# Polynomial Regression
# -------------------------------

poly = PolynomialFeatures(degree=2)
```

```python
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)

poly_pred = poly_model.predict(X_test_poly)

poly_mse = mean_squared_error(y_test, poly_pred)
poly_r2 = r2_score(y_test, poly_pred)

# -------------------------------
# Comparison
# -------------------------------

print("Linear vs Polynomial Regression")
print("---------------------------------")

print("\nLinear Regression:")
print("Mean Squared Error:", linear_mse)
print("R2 Score:", linear_r2)

print("\nPolynomial Regression:")
print("Mean Squared Error:", poly_mse)
print("R2 Score:", poly_r2)

# -------------------------------
# Plot
# -------------------------------

plt.figure(figsize=(10, 6))

# Plot actual training and test points
plt.scatter(X_train, y_train, label="Training Data", alpha=0.6, color='blue')
plt.scatter(X_test, y_test, label="Actual Test Data", alpha=0.8, color='red')


# For plotting the regression lines, we need sorted X values to draw a smooth
curve
# Create a range of x values for plotting the models' predictions
X_plot = pd.DataFrame({'day': range(X['day'].min(), X['day'].max() + 1)}) #
Ensure all days are covered

# Linear Regression predictions for plotting
linear_pred_plot = linear_model.predict(X_plot)
plt.plot(X_plot, linear_pred_plot, label="Linear Regression Fit",
color='green', linestyle='--')

# Polynomial Regression predictions for plotting
X_plot_poly = poly.transform(X_plot)
poly_pred_plot = poly_model.predict(X_plot_poly)
plt.plot(X_plot, poly_pred_plot, label="Polynomial Regression Fit (degree=2)",
color='purple', linestyle='-.')
```

```
plt.xlabel("Day (Numerical)")
plt.ylabel("Play (0=No, 1=Yes)")
plt.title("Linear vs Polynomial Regression for 'Day' vs 'Play'")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT:

```
Linear vs Polynomial Regression
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Linear Regression:
Mean Squared Error: 0.22827631542691593
R2 Score: -0.027243419421121517

Polynomial Regression:
Mean Squared Error: 0.10688512797655085
R2 Score: 0.5190169241055212
```